Faculty of engineering, Cairo University
Electronics and Electrical Communication Department
ELC4046 – Process Control

Lab 9 _ Tuning From Open Loop Tests

ELC 4046 | Fall 2026

Submitted to :

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LABORATORY EXERCISE 9
TUNING FROM OPEN LOOP TESTS

OBJECTIVE: To provide practice in open-loop testing, estimation of process parameters and calculation of tuning parameters from the open loop test data.

PREREQUISITE: Completion of Exercises

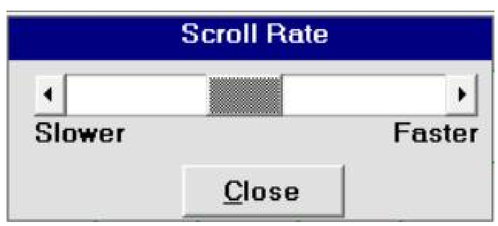
1	Process Dynamic Characteristics
8	PID Controller Characteristics
25	Modifications to The Textbook PID

or equivalent level of instruction or experience.

1. RUNNING THE PROGRAM

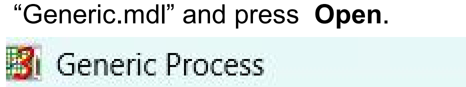
Run **PC-ControLAB**.

Slow down the simulation speed via **View | Scroll Rate** and select a mid-value.



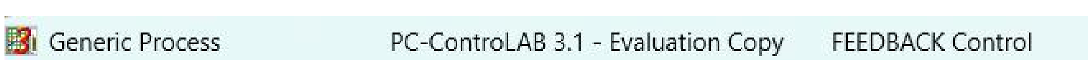
If PC-ControLAB is already running, then reread the “GENERIC” process model to initialize the program:

From the menu bar, select **Process | Select Model**.
Highlight “Generic.mdl” and press **Open**.



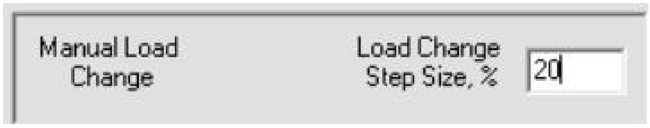
Confirm the following:

Process:	GENERIC	(see the top line, left hand side)
Control Strategy	FEEDBACK	(see the top line, right hand side)

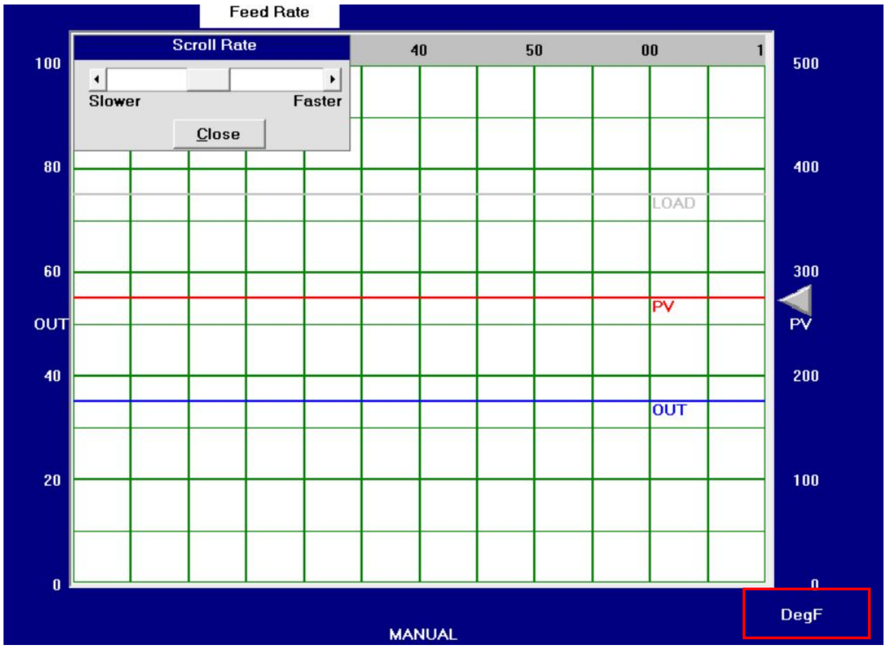


OPTIONAL: If you are more familiar with “Proportional Band” rather than “Gain”, or the reset setting in “Repeats/Minute” rather than “Minutes/Repeat”, then go to **Tune | Options** tab and choose the settings that you are more familiar with.

Select **Load** and set Manual Load Change Step size to 20%.

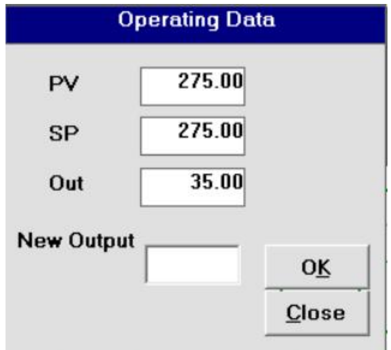


If the right-hand scale of the grid is not in engineering units, then select **View | Display Range | Engineering Units**

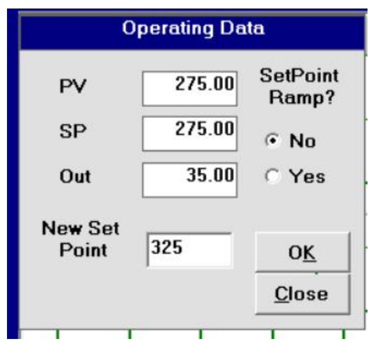


2. TRIAL-AND-ERROR TUNING

This program begins operation with a PV of 275 DegF, a controller output of 35% and a load variable (feed rate) of 300 GPM. Assume that this is the normal operating point for this process.



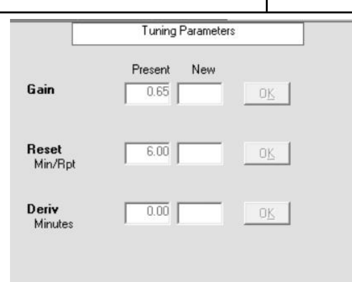
Trial-and-error tuning requires the engineer to observe the PV response to an event, usually a setpoint change, then decide what tuning parameter (or parameters) should be changed, in which direction, and by how much.



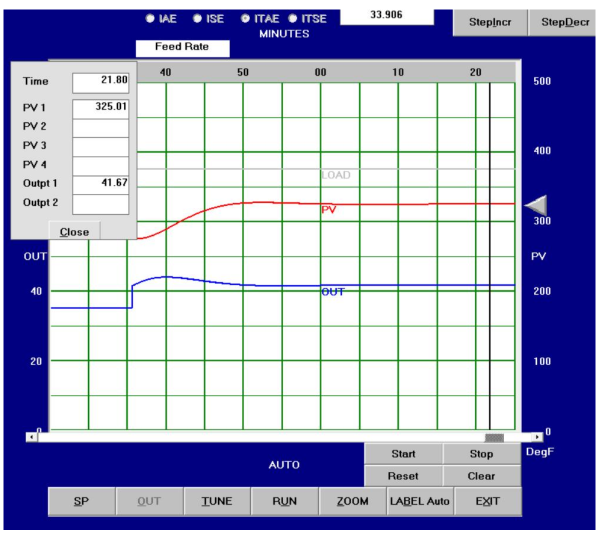
For the sake of comparison with formal tuning (to be done later in this exercise), first follow the procedure in *Basic and Advanced Regulatory Control, second edition, chapter 6, page 127*, to tune the controller by trial and error. When you are done, document your tuning parameters for the PI controller.

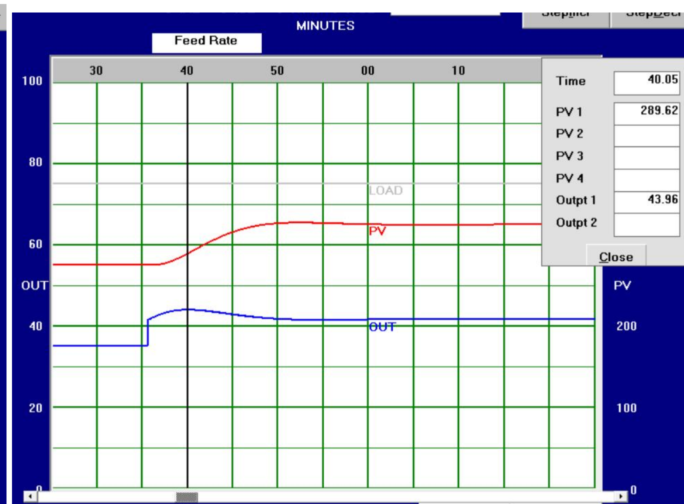
Change the setpoint, either up or down, by 50 DegF. Calculate or measure the overshoot, decay ratio, period, manipulated variable overshoot (*see Figure 2*), the settling time and ITAE. You may need to zoom in by pressing **ZOOM** and selecting appropriate values. We adjusted the tuning parameters iteratively and evaluated several parameter combinations before achieving a satisfactory response. After multiple adjustments , a good compromise between response speed, stability, and overall control performance was obtained.

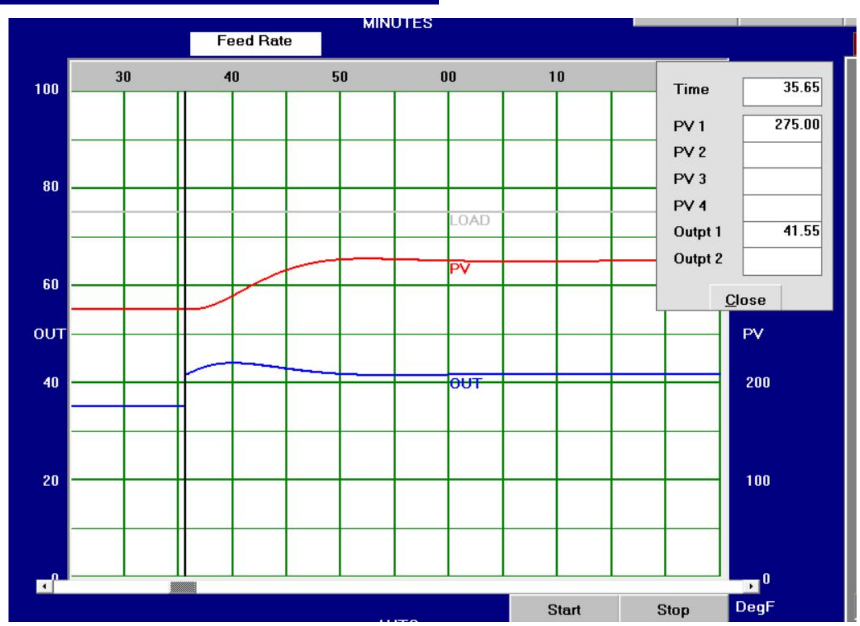
Gain (K_C)	0.65
Reset Time (T_I) (min/rpt)	6



Setpoint change	50
Settling time	48-35.65=12.35min
Overshoot (%)	4.42%
Decay Ratio	0
Period / T_I	-
Manipulated variable overshoot (%)	B=6.66,a=2.28 A/b=34.234%
ITAE (Select **View | Display Performance Meas** to setup)	33.906

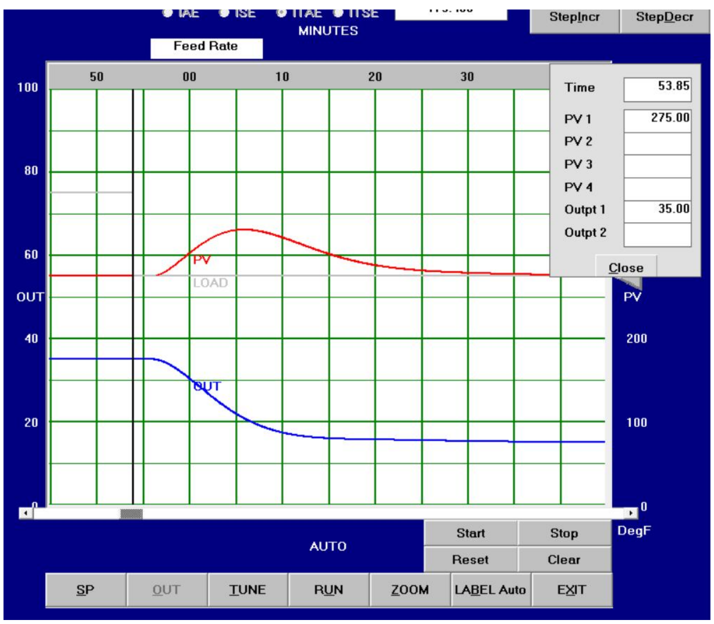


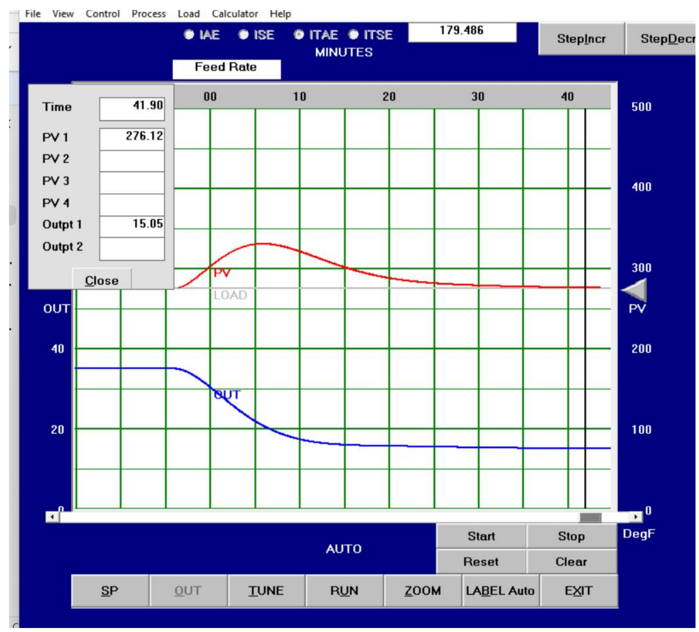




Further, test the controller by making 20% load change. Press **StepIncr** or **StepDecr** once. Record the relevant data in the following table.

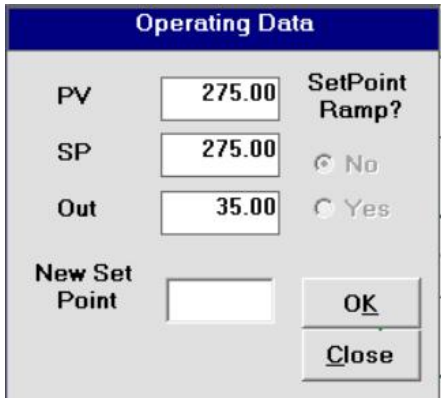
Load upset	**20%**
Peak deviation from setpoint (DegF)	330.84-275=55.84
Settling time (*2% remaining of the no-feedback steady-state deviation*)	48.05min
Decay Ratio	-
Manipulated variable overshoot (%)	-
ITAE	179.486



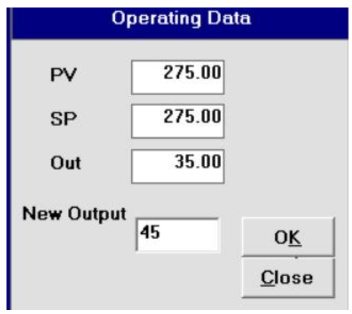


3. IDENTIFICATION FROM OPEN LOOP DATA

Return to the normal operating point of 275 DegF, controller output of 35% and a load variable of 300 GPM.



With the controller in MAN, change the output to 45%. (On a real process, you may not be able to make that much change in controller output.)

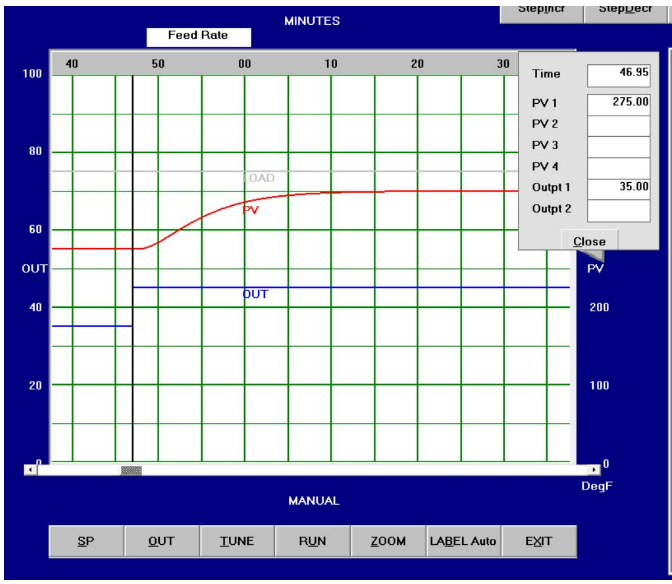


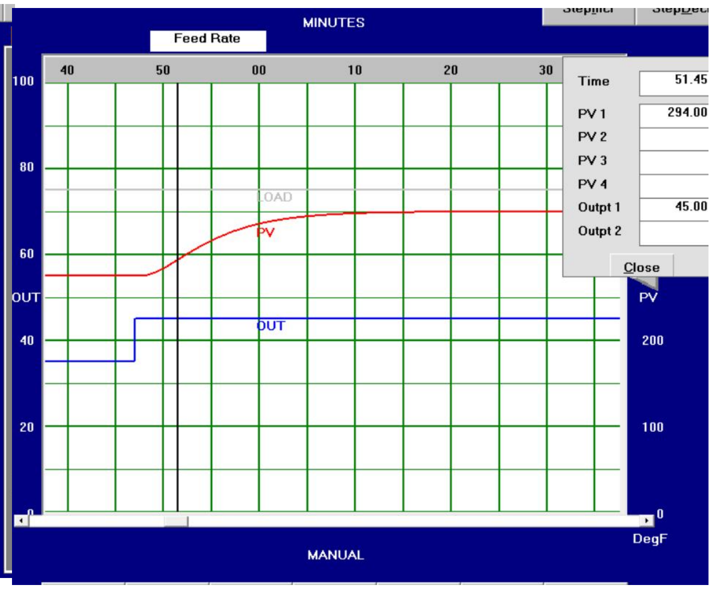
Estimate:
(see Figure 1 at the back of this exercise on how to estimate process parameters.)

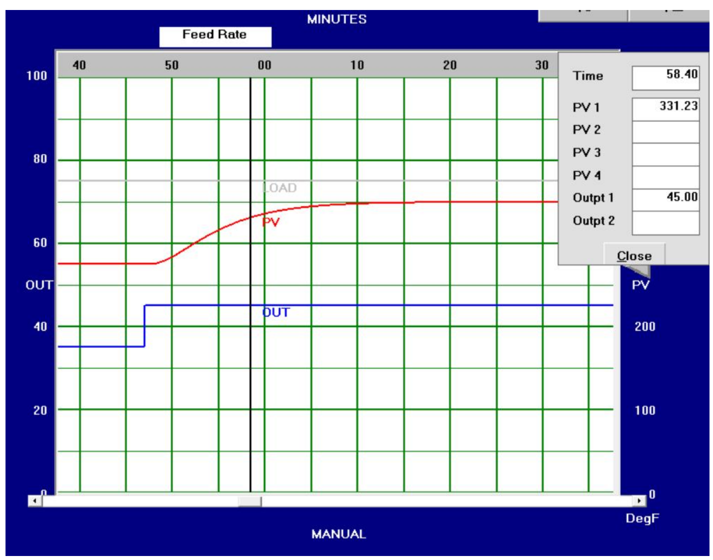
Process gain (K_P) $\dfrac{\text{\% change in PV}}{\text{\% change in controller output}}$ ___1.5_____________

Dead time (θ) ___2.6__________ min

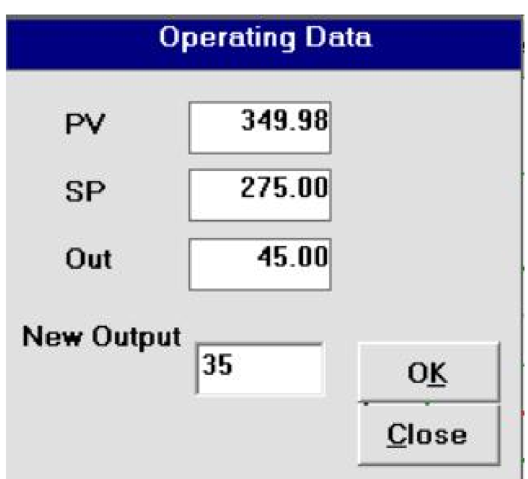
Time constant (τ) ___6.32_________ min







Return the controller output to 35%.



In a later section, you will have the chance to try a more rigorous process identification method.

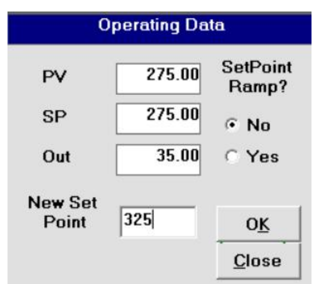
4. TUNING AND TESTING THE FEEDBACK LOOP RESPONSE

Calculate tuning parameters for the PI controller, for each algorithm below. Enter them in the following table.

	Heuristic	**Ziegler-Nichols**	**Lambda tuning** $\lambda = 4.46$	**Myke King P on PV**
Gain (K_C)	1.33	1.46	0.5968	0.774
Reset Time (T_I) (min/rpt)	8.92	8.66	6.32	5.369

NOTES:
(1) With the controller in MAN, select **Control | Control Options,** and select PID Non-interacting control algorithm.
(2) For the P on PV algorithm ONLY, remember to select **Control | Control Options,** and select Prop on Meas.

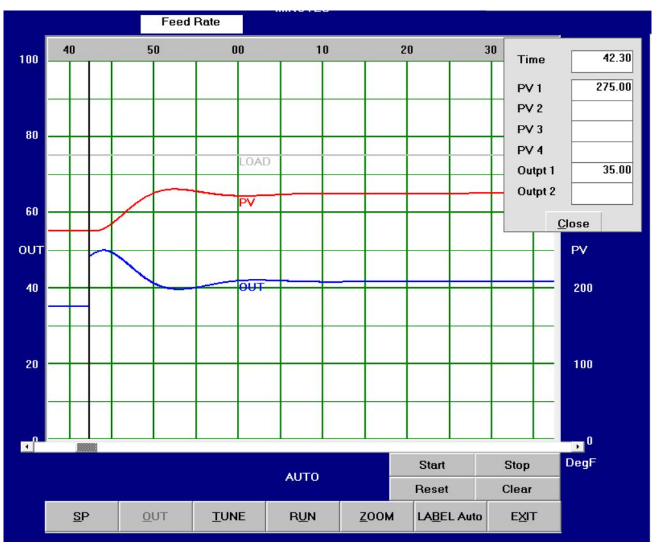
For each tuning method, enter the parameters above (and, naturally, Deriv Time = 0), put the controller in AUTO and make a setpoint change. (Change the setpoint, either up or down, by 50 DegF. On the job, you probably cannot make that large of change.)

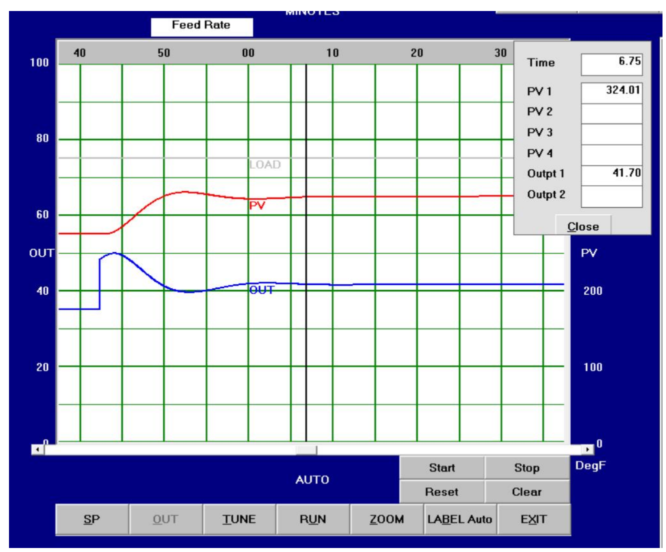


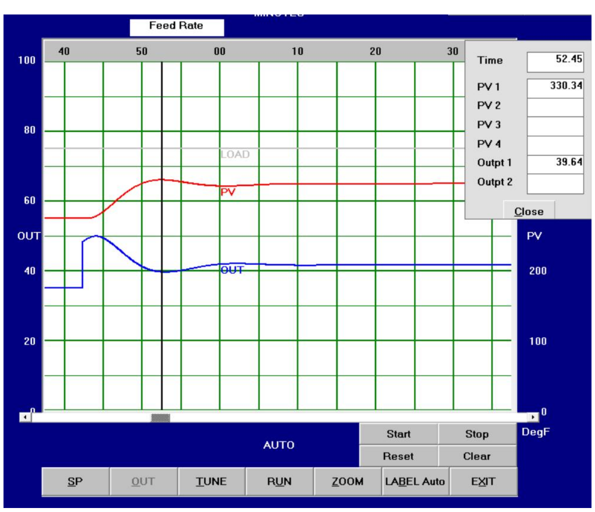
Fill in the table below. You may need to zoom in by pressing **ZOOM** and selecting appropriate values.

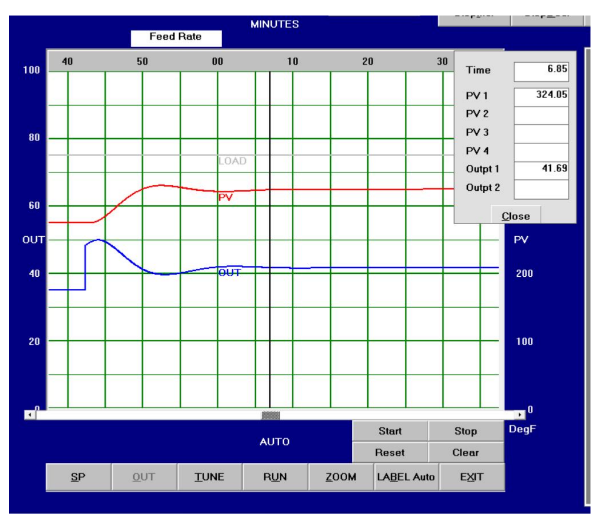
Setpoint change	Heuristic	Zeigler-Nichols	Lambda tuning	Myke King P on PV
Settling time	24.45 min	23min	14.75min	18.3min
Overshoot (%)	10.68%	14.96%	0.9%	0.74%
Decay Ratio	0	0.02	0	0
Period	14.4min	17.2min	-	-
Period / T_I	1.6143	1.98	-	-
Manipulated variable overshoot (%)	123.874%	145.645%	23.98%	18.5%
ITAE	26.737	26.301	35.691	50.256
Settling time / $(\theta+\tau)$	2.741	2.5785	1.65	2.05

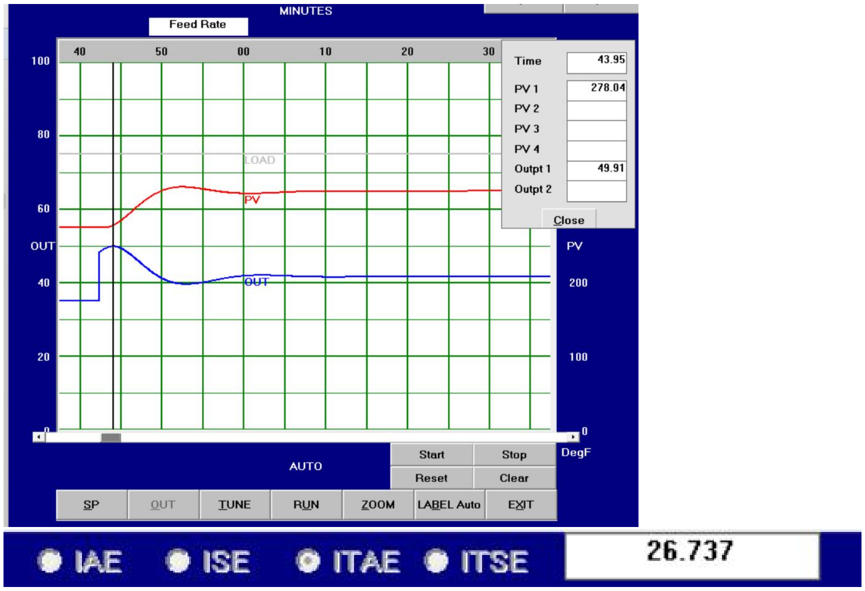
1) Heuristic



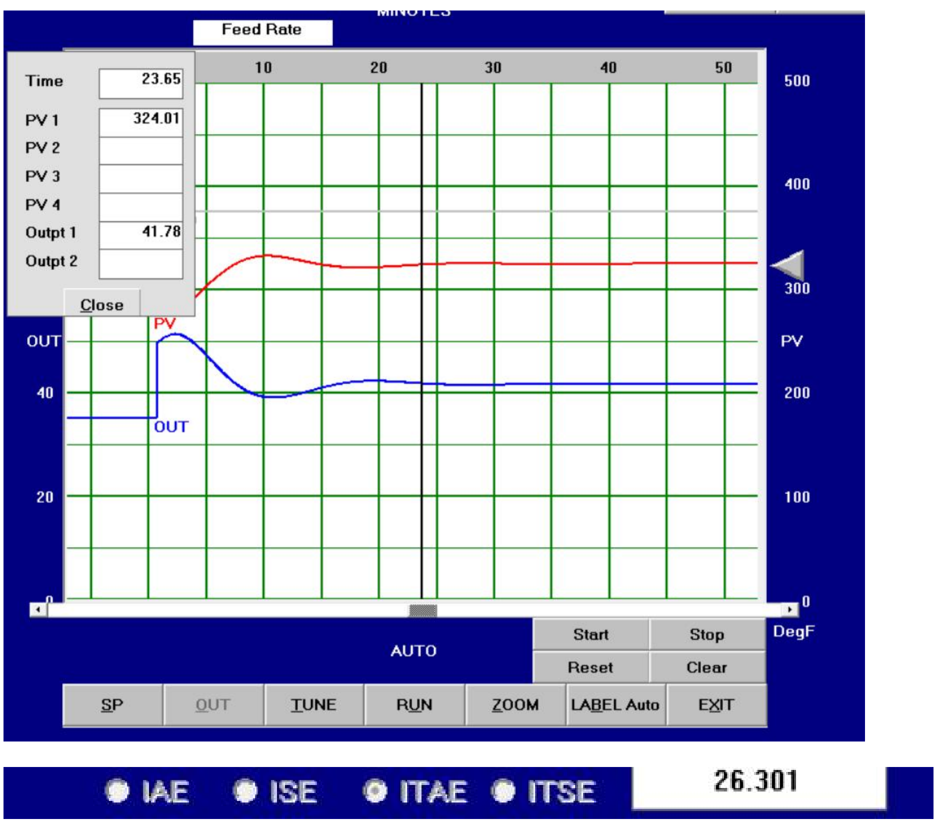




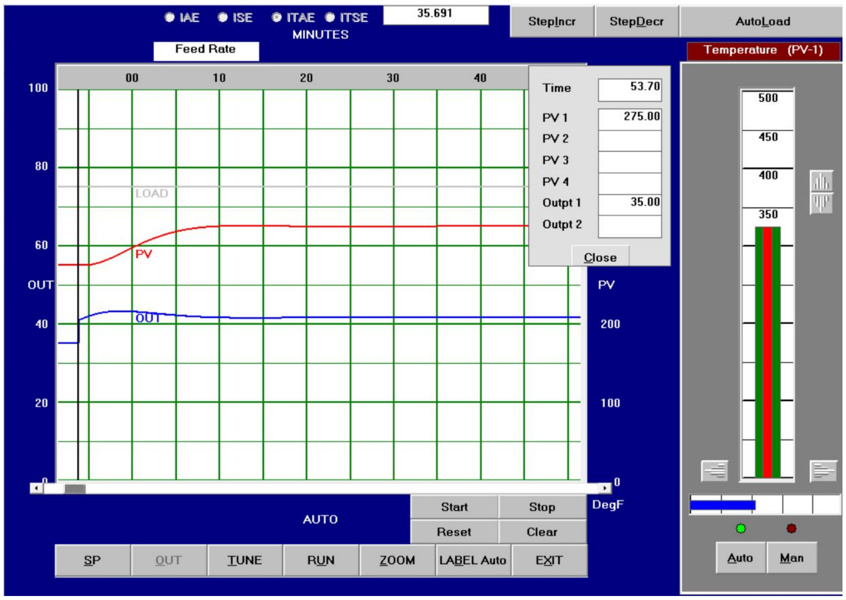




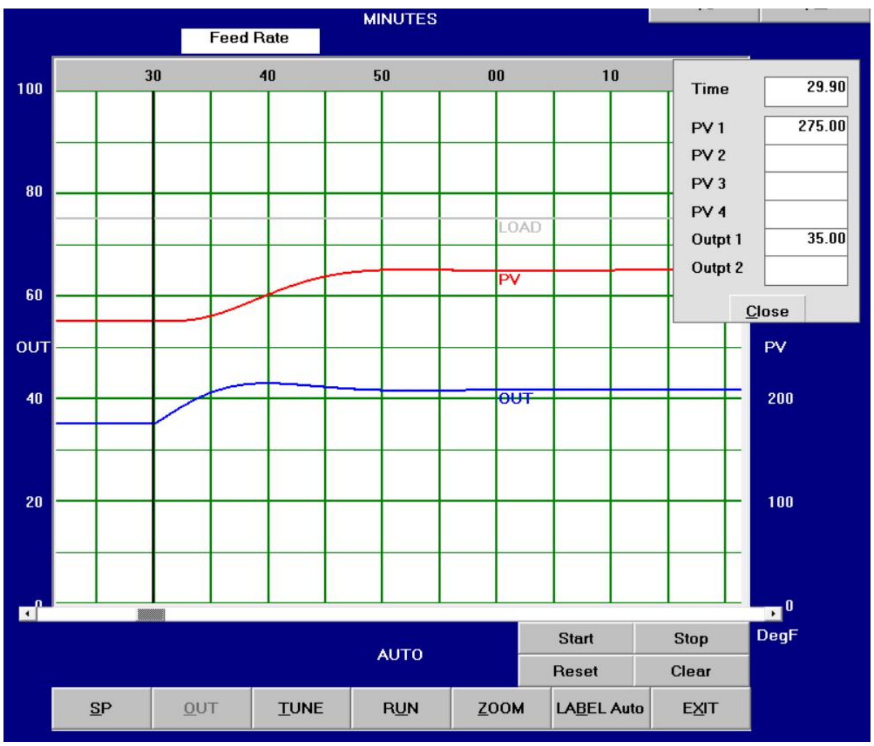
2) Zeigler-Nichols



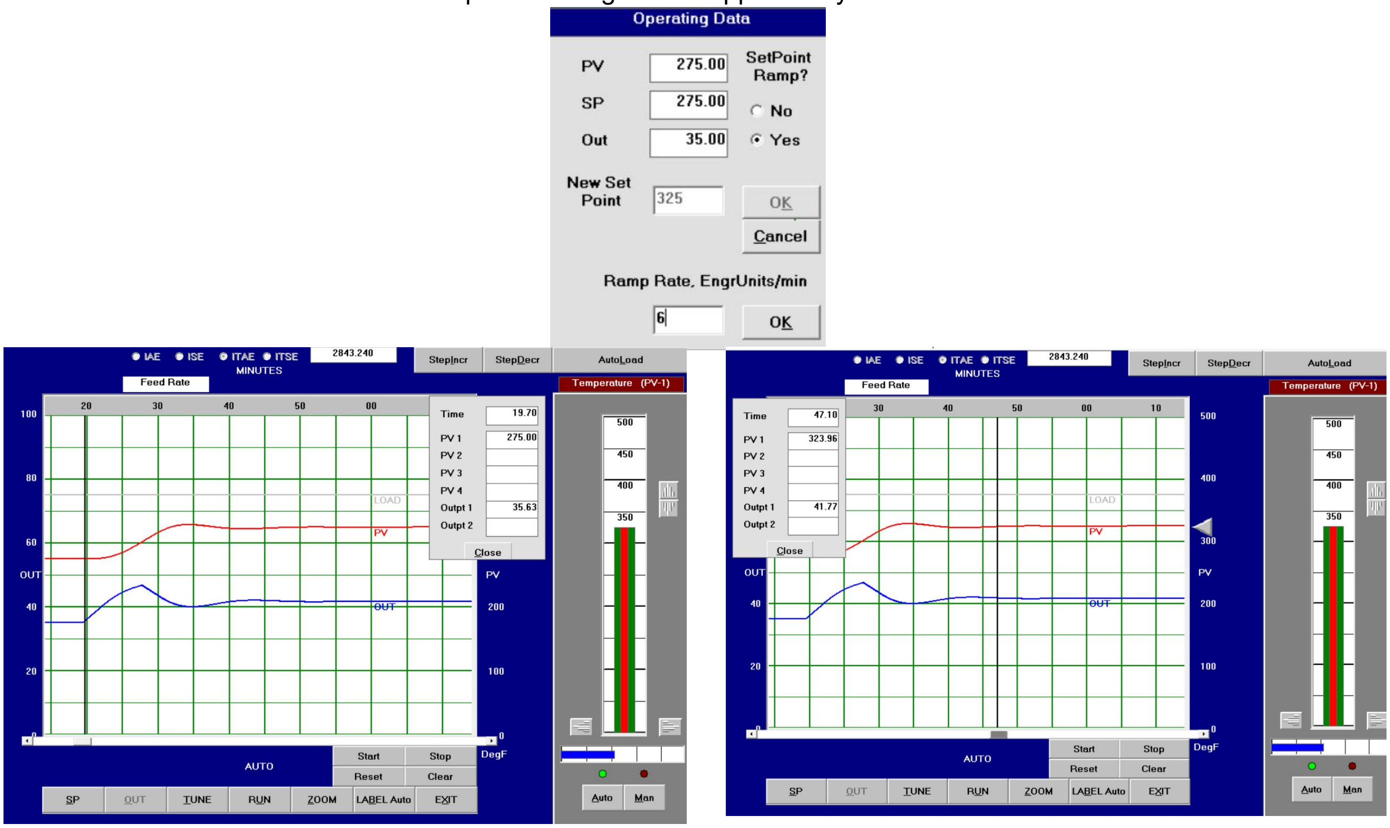
3) Lambda



4) Myke king



Repeat the above test, enabling the **SetPoint Ramp** option (enter a rate from 3 to 9 DegF/min, tweak this value as required) before making the same setpoint change. We use setpoint ramping since the more common setpoint filtering is not supported by this version of the software.



Setpoint change	Zeigler-Nichols + setpoint ramping Ramp rate =6
Settling time	27.4min
Overshoot (%)	7.5%
Decay Ratio	0
Period	No period
Period / T_I	no
Manipulated variable overshoot (%)	75.8%
ITAE	2843.240
Settling time / $(\theta+\tau)$	3.07

Does the Lambda tuning response roughly appear as an exponential one with some delay and a time constant of l?
Correct, the response approximately behaves like an exponential one with a delay and a time constant of λ.

For the P on PV algorithm, does the smoother manipulated variable movement at the onset of the setpoint change degrade its performance compared to the other algorithms?
The smoother manipulated variable movement in the P on PV algorithm does not mean worse performance. Since there is no proportional kick, the controller response is less aggressive during setpoint changes. Although the initial response may be slightly slower, it results in smoother operation and lower overshoot. According to the results, P on PV achieved the lowest process overshoot and a low MV overshoot compared to the other algorithms.

Which responses can be described as well-damped and which ones can be concluded as a bit-overdamped?

Lambda and Myke King responses can be described as well-damped, while none of the responses show clearly overdamped behavior.

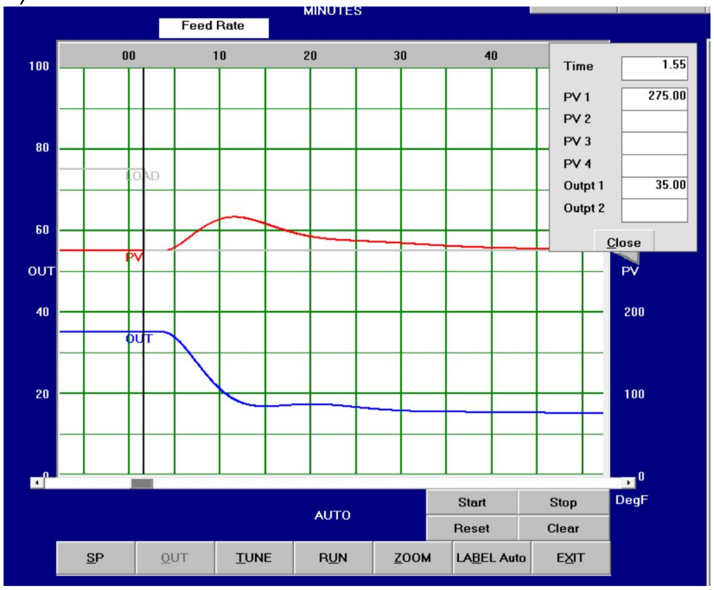
Which responses amongst the well-damped ones are inferior when conclusive measures are considered?
Myke King response which has the largest ITAE

Does the response with minimum ITAE indeed correspond to the smallest settling time?
No, the response with the minimum ITAE does not have the smallest settling time

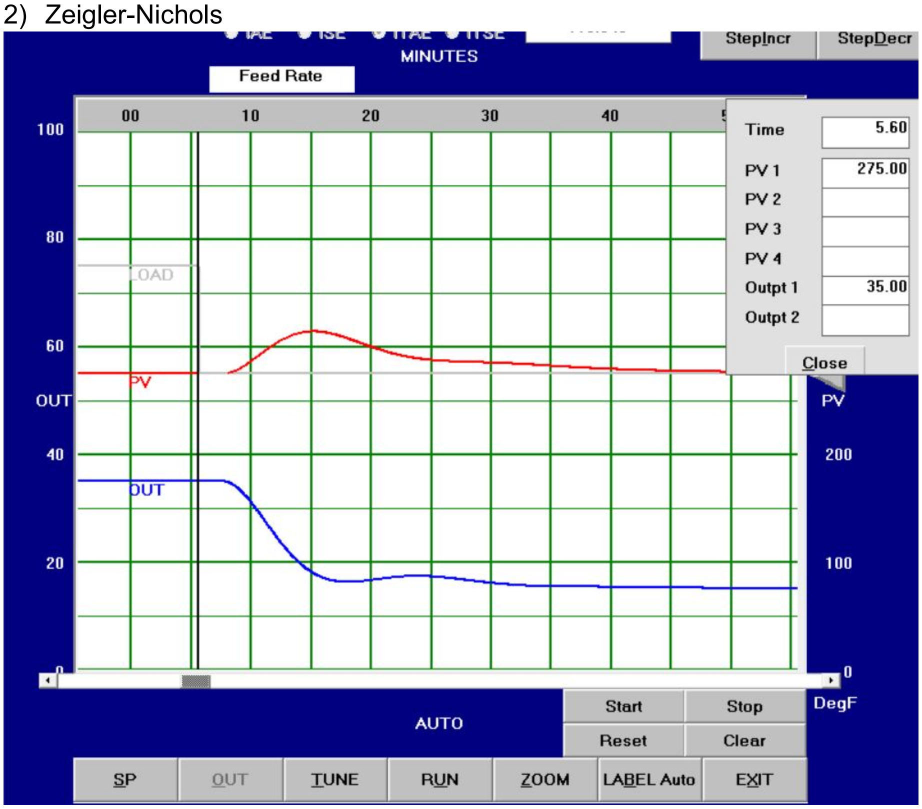
Further, test each controller by making 20% load change. Press **StepIncr** or **StepDecr** once**.** Record the relevant data in the following table.

Load upset	**Heuristic**	**Zeigler-Nichols**	**Lambda tuning**	**Myke King P on PV**
Peak deviation from setpoint (DegF)	316.5-275=41.5	314.38-275=39.38	333.4-275=58.4	325.81-275=50.81
Settling time	55.8	53.05	51.7	43.7
Decay Ratio	0	0	0	0
Manipulated variable overshoot (%)	8.6%	6.35%	-	-
ITAE	67.059	94.315	120.251	69.229

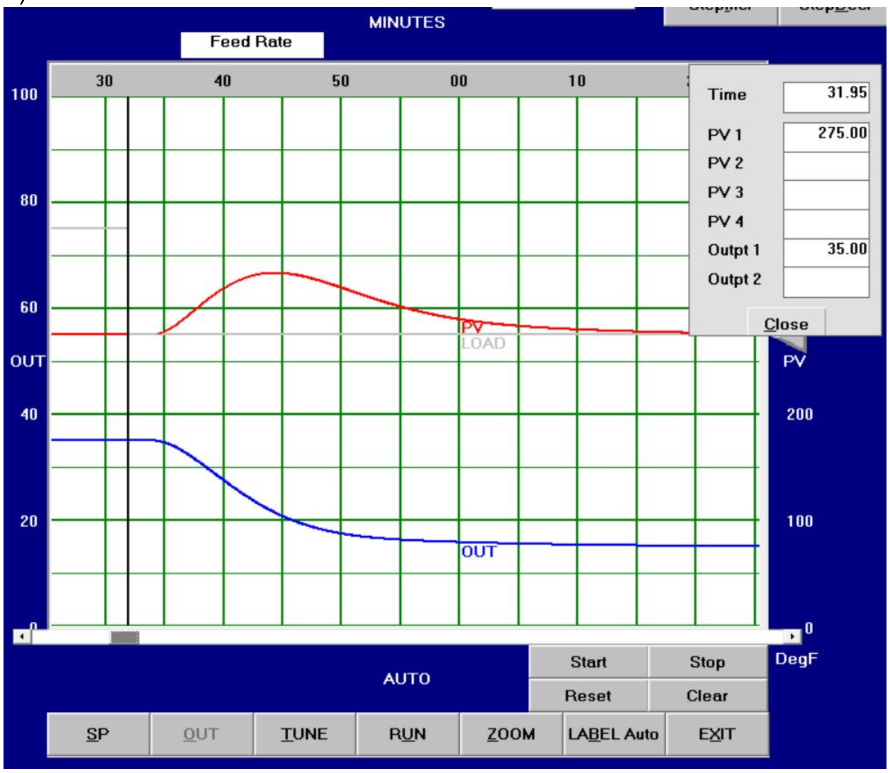
1) Heuristic



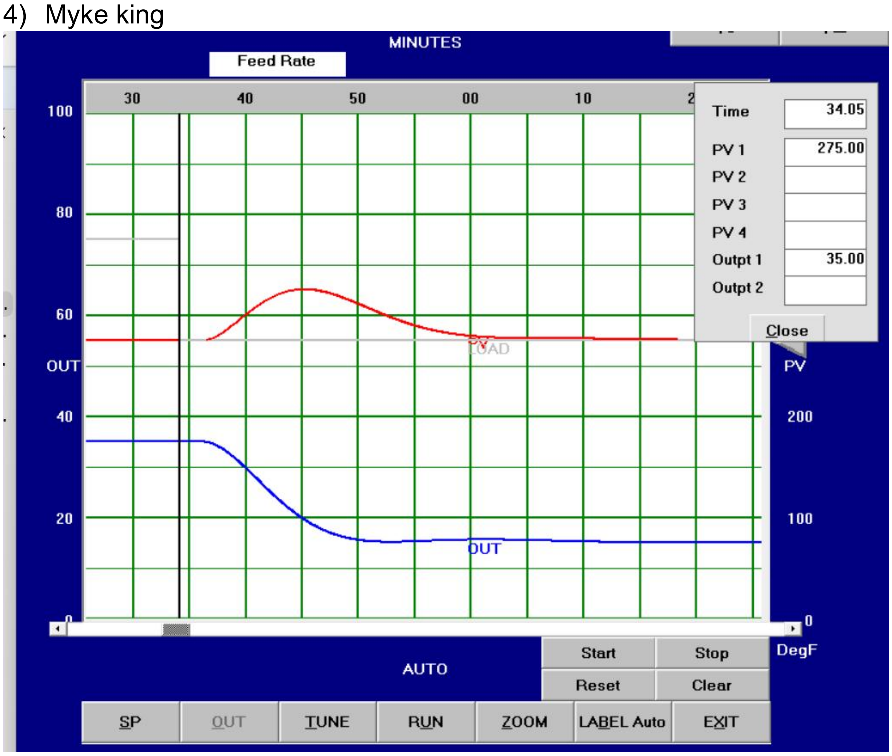
2) Zeigler-Nichols



3) Lambda



4) Myke king



Does Zeigler-Nichols method achieve quarter cycle decay as it aims for? __No________

Should we redo the load upset test above with setpoint ramping enabled? ___No_______

Why?

_In a load upset test, the setpoint remains constant and the disturbance is applied to the PV instead.___

Based on your findings above, which tuning method would you use?

Based on the findings, the Zeigler–Nichols method would be preferred for load change tests if we need small peak deviation. However, if strong performance is required for both setpoint and load changes, the Myke King method would be a better choice .

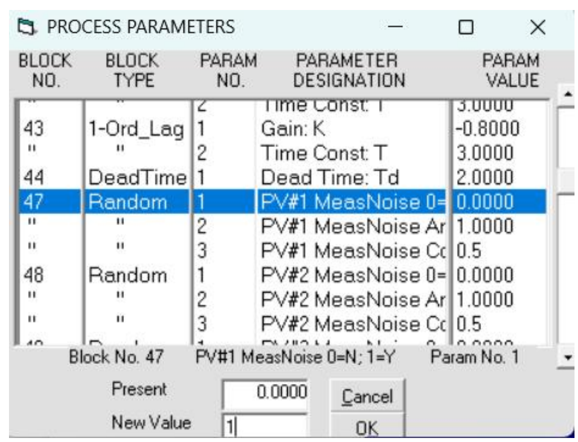
How does trial-and-error tuning in section 2 compare to formal tuning, in time, effort, and results?

Compared to trial-and-error tuning, formal tuning required less time and less manual effort because the controller parameters were obtained using established methods rather than repeated adjustments. In terms of results, formal tuning achieved better and more consistent control performance, while trial-and-error tuning relied more on experimentation and produced less optimized responses.

5. EFFECTS OF A NOISY PROCESS VARIABLE

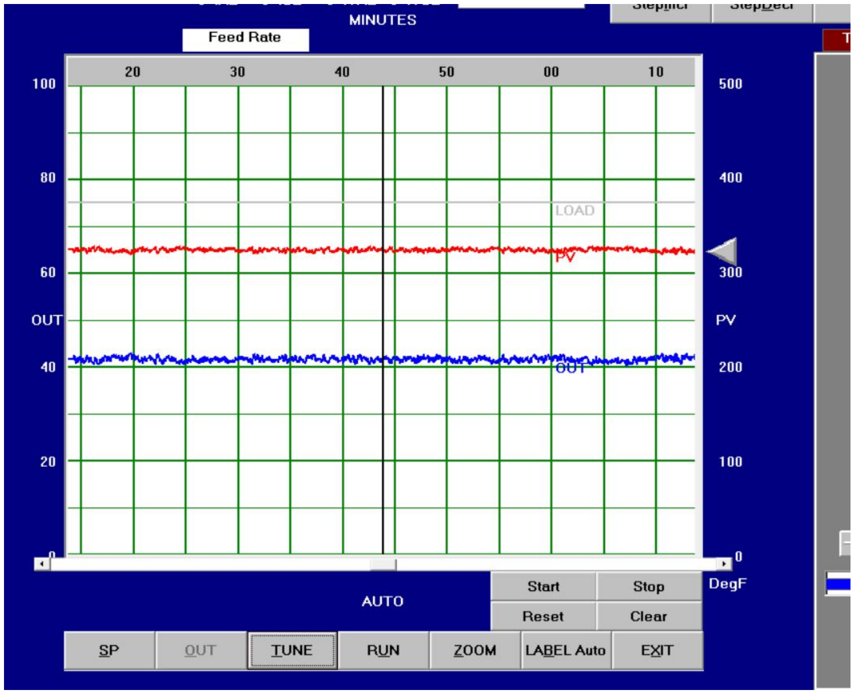
Note, what you have seen so far is a perfectly noise-free process. Real processes usually have some noise on the measurement. With the controller in AUTO:

Select **Process | Change Parameters**
Use the scroll bar to scroll down until you see
"PV#1 Meas Noise 0-N; 1-Y"
Highlight that, then enter "1"
Press **OK** then **Close**



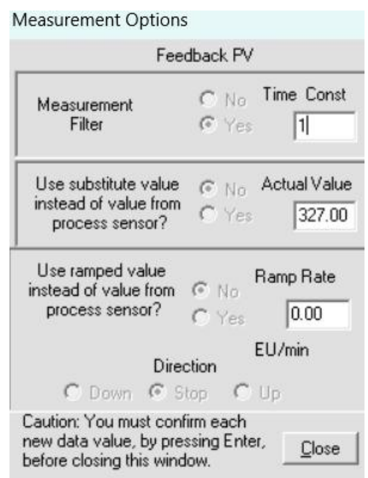
Describe what you see:

Noise was added to both the PV and the MV signals



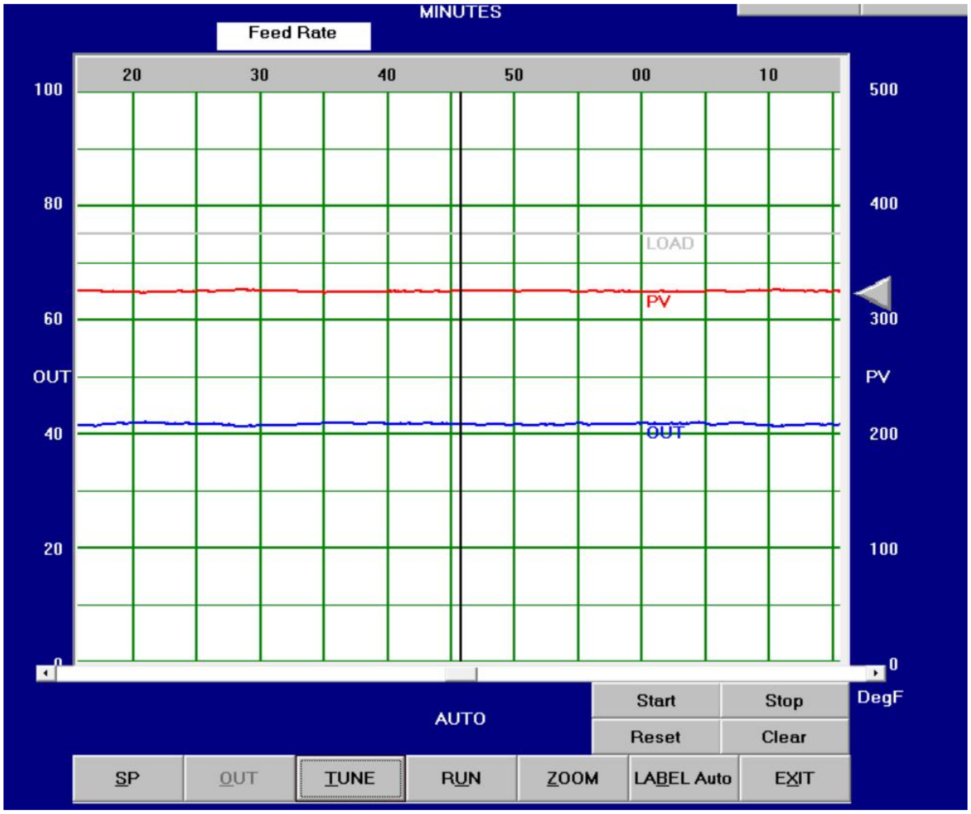
Frequently the measurement signal is filtered to "hide" the effect of the noise. (The noise is still there - you just can't see it!) We will put a filter on the measurement. This is equivalent to implementing a software filter on the analog input block on a DCS. EXCEPT: Due to the relatively slow sampling rate of this simulation, as compared with the typical sampling rate of a DCS, we will use a larger filter time constant here than would be used in real life.

Select **Control | Measurement Options** in the drop-down menu
Click in the circle "Yes" in the section "Measurement Filter"
Enter "1.0" (minutes) for the filter time constant
Press **Close**


Measurement Options
Feedback PV
Measurement Filter — No / Yes — Time Const: 1
Use substitute value instead of value from process sensor? — No / Yes — Actual Value: 327.00
Use ramped value instead of value from process sensor? — No / Yes — Ramp Rate: 0.00 EU/min
Direction: Down / Stop / Up
Caution: You must confirm each new data value, by pressing Enter, before closing this window.
Close

Describe the effect:

After applying the filter, the noise levels in both the PV and MV signals were significantly reduced compared to the unfiltered case.



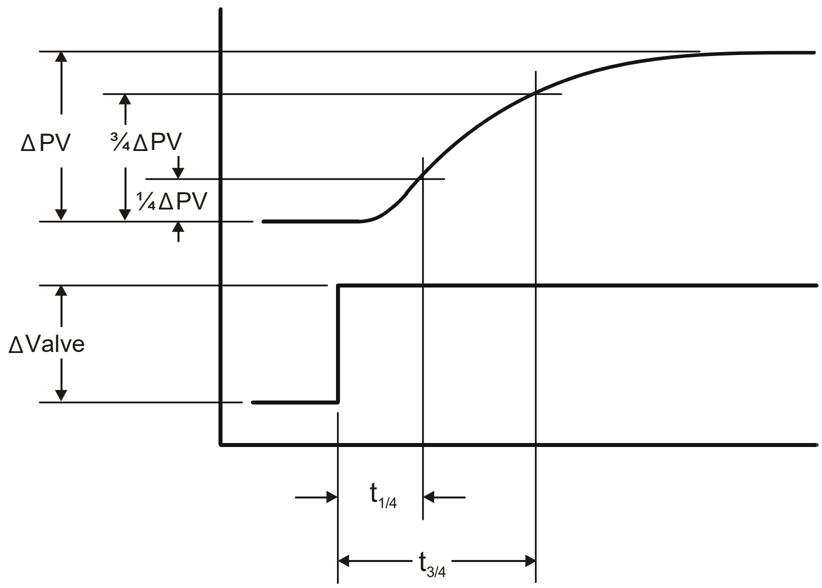


Figure 1 DETERMINING PROCESS PARAMETERS

Process gain: $K_P = \dfrac{\Delta PV \,/\, PV\ range}{\Delta VALVE \,/\, MV\ range}$

Measure the time from the instant of valve change until the process rises by 1/4 of its total amount. (This obviously must be done from a chart record after the process has reached its new equilibrium.) Call this time $t_{1/4}$.

Measure the time from the instant of valve change until the process rises to 3/4 of its final amount. Call this time $t_{3/4}$.

When these two times are determined, calculate the dead time and process time constant from the following equations:

$$\tau = 0.91(t_{3/4} - t_{1/4})$$

$$\theta = t_{3/4} - 1.4\tau$$

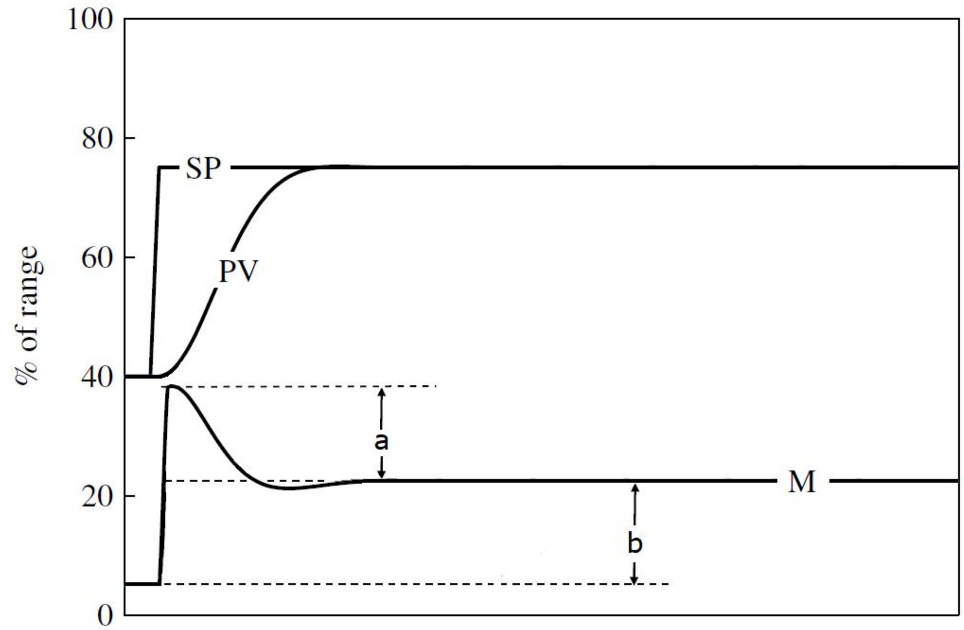


Figure 2 DETERMINING MANIPULATED VARIABLE OVERSHOOT

MV overshoot = a/b %